

MCU (STM32WB55), SWD Debug & RF (BLE) — BLEC-SCH-0002 Non-medical wearable prototype

Essential performance (controller role):

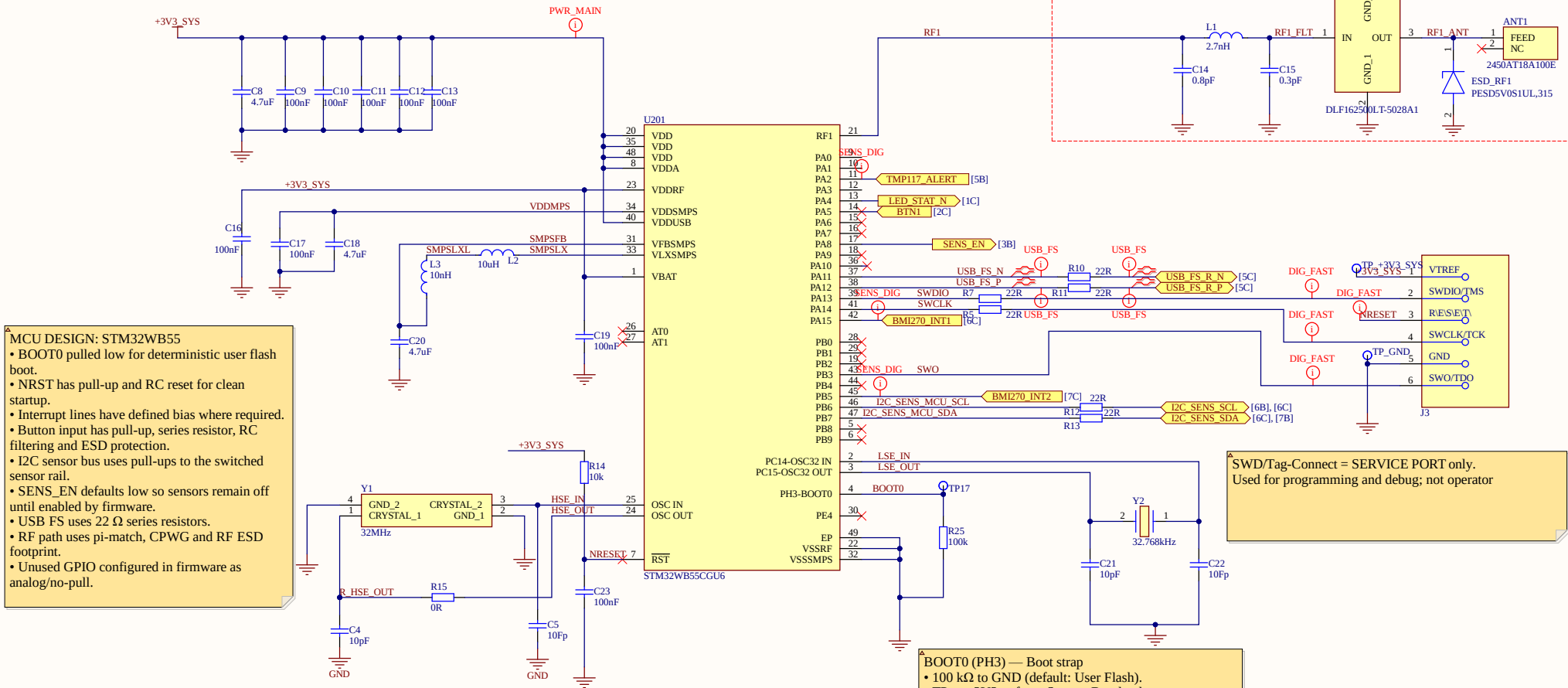
- Board provides BLE command/control only; no direct patient-applied parts.
- All MCU I/Os are SELV and default to safe states on reset (pull-ups/-downs and series resistors as shown).
- BLE link integrity and stimulation safety are handled at system/implant level.

IEC 60601 Compliance:

USB-C = AC/DC SELV Input Port.
PPTC + TVS + ESD + CMC provide protection for IEC 61000-4-2/4-4/4-5.
All voltages SELV; battery safety provided by charger + pack.

RF port (intentional radiator):

2.4 GHz BLE antenna with π -match (C-L-C) and RF ESD footprint.
CPWG, via fence and π -match used to meet radiated emission/immunity requirements (IEC 60601-1-2).



MCU DESIGN: STM32WB55

- BOOT0 pulled low for deterministic user flash boot.
- NRST has pull-up and RC reset for clean startup.
- Interrupt lines have defined bias where required.
- Button input has pull-up, series resistor, RC filtering and ESD protection.
- I2C sensor bus uses pull-ups to the switched sensor rail.
- SENS_EN defaults low so sensors remain off until enabled by firmware.
- USB FS uses 22 Ω series resistors.
- RF path uses π -match, CPWG and RF ESD footprint.
- Unused GPIO configured in firmware as analog/no-pull.

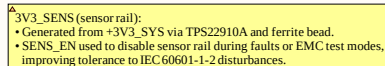
SWD/Tag-Connect = SERVICE PORT only.
Used for programming and debug; not operator

- BOOT0 (PH3) — Boot strap
- 100 k Ω to GND (default: User Flash).
- TP to +3V3 to force System Bootloader.
- Do not use as GPIO.

Title BLE-Control Wearable — Schematic Set (AD25)

Size	Number	Revision
A3	BLEC-SCH-0002	A0 (EVT)
Date:	5/26/2026	Sheet of 2 of 3
File:	C:\Users\MCU_RF\SchDoc	Drawn By: Caoilte Donohoe

Sensor Power Gate



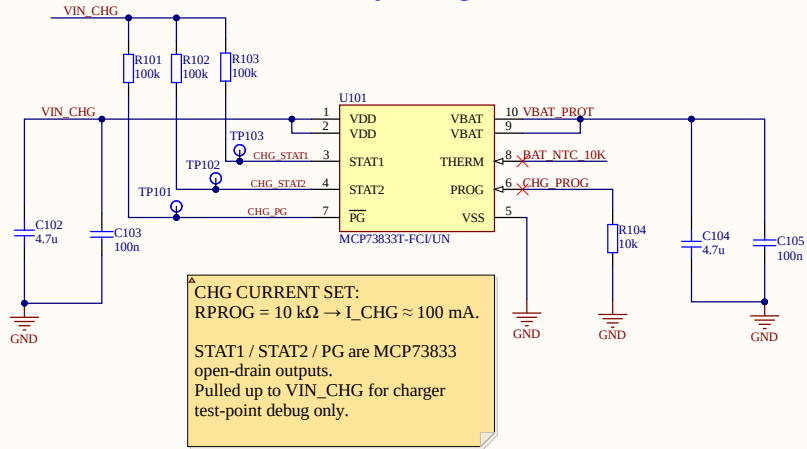
- Operator I/O (button + LED + skin sensor):
 - Button: TVS at pad + 100R series + RC network for ESD and burst immunity.
 - Sensors: on-board only, short I2C, powered via 3V3_SENS gate.
 - No direct patient-applied parts on this PCB; temperature is read through enclosure/assembly.
- BTn1 default state:
 - Pull network ensures defined logic level on reset and under noise.



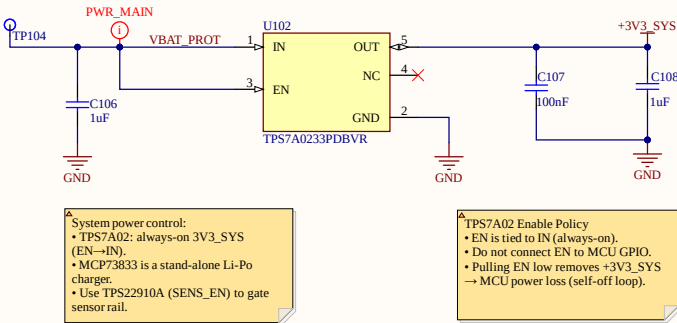
Title BLE-Control Wearable — Schematic Set (AD25)			
Size A3	Number BLEC-SCH-0003	Revision A0 (EVT)	
Date: 5/26/2026	Sheet of 3 of 3		Drawn By: Caitie Donohoe
File: C:\Users\LED_Sch\Documents\BLEC-SCH-0003\BLEC-SCH-0003_Schematic_Set.dwg			

Power, Battery, USB-C & MCP73833T (Main 3V3) — BLEC-SCH-0001 Non-medical wearable prototype

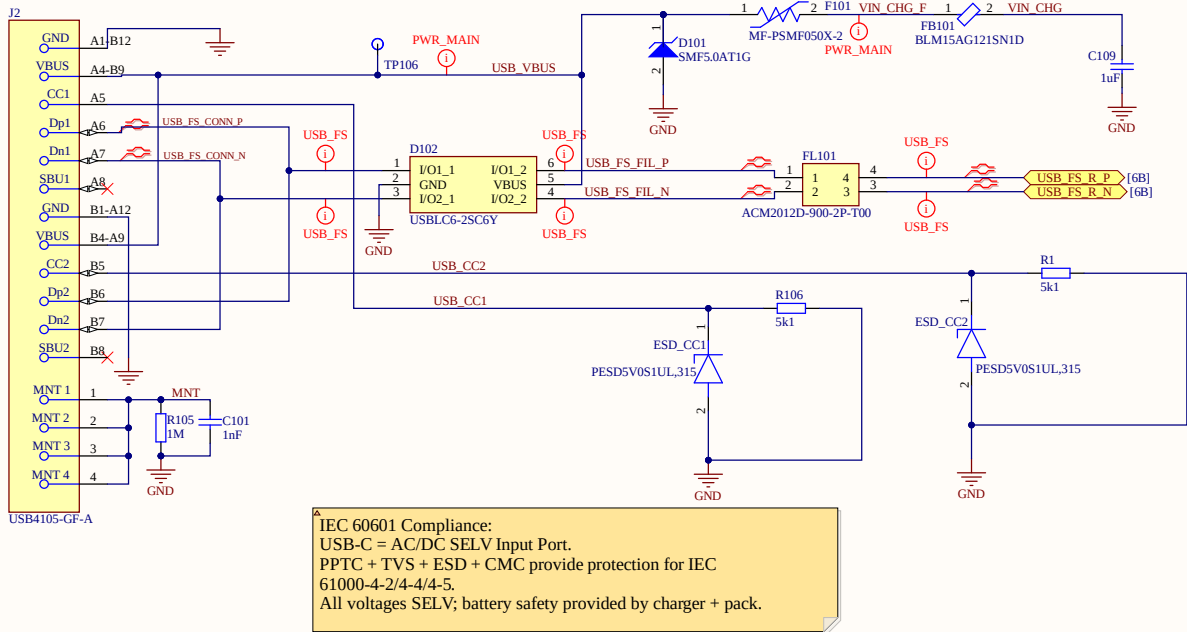
Battery Charge Cct



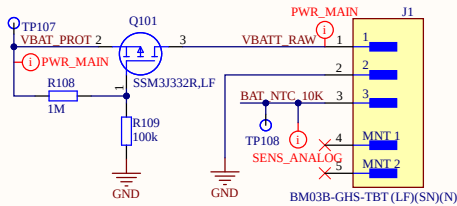
REG LINEAR



USB Filtering and ESD



Battery Connect



Title			BLE-Control Wearable — Schematic Set (AD25)	
Size	Number	Revision		
A3	BLEC-SCH-0001	A0 (EVT)		
Date:	5/26/2026	Sheet of 1 of 3		
File:	C:\Users\Power_Charge_USB\SchDoc	Drawn By: Caitie Donohoe		